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Can AI Models Summarize Your Diary Entries? Investigating Utility of Abstractive Summarization for Autobiographical Text

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ABSTRACT

Journaling is a widely adopted technique, known to improve mental health and well-being by enabling reflection on past events. Large amounts of text in digital journaling applications could hinder the reflection process due to information overload. Abstractive summarization can solve this problem by generating short summaries to quickly glance at and reminisce. In this paper, we present an investigation of the utility of large language models in the context of autobiographical text summarization. We study two approaches to adapt a self-supervised learning (SSL) model to the domain of autobiographical text. One model employs transfer learning using our new autobiographical text summary dataset to fine-tune the SSL model. The second model leverages existing news datasets for high-quality text summarization mixed with our autobiographical summary dataset. We conducted mixed methods research to analyze the performance of these two models. Through objective evaluation using ROUGE and BART scores, we find that both these approaches perform significantly better than the SSL model fine-tuned with only high-quality news datasets, showing the importance of domain adaptation and autobiographical text summary dataset for this task. Secondly, through a subjective evaluation on a crowd-sourcing platform, we evaluated the summaries generated from these models on various quality criteria such as grammar, non-redundancy, structure, and coherence. We found that on all criteria, these summaries score >4 out of 5, and the two models show comparable results. We deployed a proof-of-concept web-based journaling application to assess the practical real-world implications of incorporating abstractive summarization in a digital journaling context. We found that the participants showed a high consensus that the summaries generated by the system captured the main idea of their journal entry (80% of the 75 participants gave a Likert scale rating of ≥ 5.0 out of 7.0, with the overall mean rating of 5.56 ± 1.32) while being factually correct, and they found it to be a useful feature of a journaling application. Finally, we conducted human evaluation studies to compare the quality of the summaries generated from a commercial tool ChatGPT and mixed distribution fine-tuned SSL model, and present insights into these systems in the context of autobiographical abstractive text summarization. We have made our model, dataset, and subjective evaluation questionnaire openly available to the research community.

KEYWORDS

Self supervised learning;
autobiographical text
summarization; digital
journaling

1. Introduction

Journaling provides a useful means to express inner thoughts and record experiences of past encounters allowing people to re-experience or reflect on thoughts and feelings associated with events or people (Ayobi et al., 2018; Backer et al., 2020; Isaacs et al., 2013; Kim et al., 2019b; Lindström et al., 2006; Wisniewski et al., 2016). Having access to these saved diary entries and going through them can improve one's emotional awareness (Ausman, 2019; Lindström et al., 2006; Sawhney et al., 2018; You et al., 2019). With the rapid improvement in Automatic Speech Recognition (ASR) Hannun et al. (2014) technology, most modern-day digital journaling applications are equipped with speech-to-text

features that transcribe users' voice recordings, opening doors to elaborate autobiographical text generation. For example, a 5-min-long audio entry could result in a text of 750 words (Yuan et al., 2006). The intended self-reflection process in journaling, which involves re-visiting the entries and reminiscing, is hindered by information overload as users can easily lose track of their diary entries due to lengthy text, i.e., large amounts of data (Agroudy et al., 2016; Alqahtani & Orji, 2019; Elsdén et al., 2016; Utley & Garza, 2011).

One potential solution would be to provide summaries of long diary entries so that users can easily get an overview. However, text entries in personal journals are authored from an autobiographical perspective and consist of feelings,

confessions, experiences, and thoughts (Johnson, 2011), which we call “autobiographical text”. In contrast to well-structured, factual, and grammatically correct descriptions such as news, the autobiographical text is highly influenced by personal writing styles, abbreviations, grammar mistakes, redundant information, and affective content. Therefore, existing approaches to generate summaries from structured text, such as news (Lewis et al., 2019; Raffel et al., 2020; Zhang et al., 2020) cannot be directly applied to autobiographical text due to domain-mismatch. Training a customized model for autobiographical text summarization in a fully supervised manner requires a large dataset containing real journal entries and corresponding human-written summaries. Existing datasets contain either structured text such as news and their corresponding gold standard summaries (Hermann et al., 2015; Narayan et al., 2018), or social media (e.g., Reddit) posts without gold standard summaries (Völske et al., 2017). Manually creating a very large dataset (a few hundred thousand) of autobiographical text along with accurate summaries is resource exhaustive due to the scale as well as due to privacy concerns when dealing with sensitive journal entries (Ferdous et al., 2016).

In this paper, we employ a pre-trained general-purpose generative language model “BART” (Lewis et al., 2019) as the base neural network model and extend its capabilities to work with autobiographical text by fine-tuning it. To address the problem of lack of datasets for autobiographical text summaries, we leverage publicly available Reddit descriptions and their titles to build a dataset, that we call *Reddit-title* dataset, consisting of autobiographical qualities (Ouyang et al., 2017). There are no gold standard summaries corresponding to Reddit descriptions, so we used highly abstractive Reddit titles of descriptions as a weak-supervision signal to train our model. To investigate methods to domain-adapt a model for autobiographical text summarization, we fine-tune the BART model in two primary ways: (1) directly fine-tune with our domain-specific autobiographical and unstructured Reddit-title dataset, i.e., *autobiographical-only*, which is motivated by the transfer learning literature (Cachola et al., 2020; Gururangan et al., 2020), and (2) fine-tune with a mix distribution of structured news data and unstructured Reddit data, i.e., *mix*, which is motivated by the ability of pre-trained language models of learning from multiple tasks and data distributions (Cachola et al., 2020; Raffel et al., 2020). The first approach is the standard technique for transfer learning where the knowledge learned by the pre-trained source model, i.e., BART model, can be adapted for the target dataset, i.e., autobiographical text summaries. The second approach incorporates ideas from scaffold tasks for multitask learning with a mix of distributions and control codes in conditional language generation to address the problem of data scarcity in the autobiographical text summary domain (Cachola et al., 2020). Specifically, we use titles from one data distribution (Reddit-titles dataset) as an auxiliary training signal by training the model to generate both titles and expert summaries from another data distribution (news dataset) indicated by control codes. The reason for mixing these two data

distributions is to learn the autobiographical qualities from the titles while the structured expert-written high-quality nature of summaries from the news dataset. We validate the quality of the summaries generated from our fine-tuned models through technical evaluation metrics as well as human evaluation studies on a crowd-sourcing platform.

We then investigate the practical implications of such an autobiographical text summarization model. For this, we develop a proof-of-concept digital journaling web application where we incorporate a summary-generating feature enabled by one of our fine-tuned models to analyze the utility of such a model in a real-world context. In addition, we compare our work against a commercial product (ChatGPT¹). Implementation of our model consisting of the fine-tuning mechanism, pre-trained model weights, as well as the web application has been made publicly available to the research community². Moreover, we compiled a comprehensive questionnaire for human evaluation of abstractive text summarization based on past literature, which is made available to the research community (see [Appendix Table A1](#)).

In summary, the aims of this work are:

- To extend the capabilities of an open source general-purpose generative language model to work with “autobiographical text” by incorporating a weak supervised signal, which needs:
 - a. A new autobiographical text dataset
 - b. A domain adaptation method for autobiographical text summarization,
- To investigate the practical implications of autobiographical text summarization in the context of a user application, as well as in comparison to a commercial product such as ChatGPT.

2. Related work

2.1. The importance of summarization of autobiographical text

Journaling involves text entries of autobiographical nature. Previous works (Ayobi et al., 2018; Elsdén et al., 2016) have discussed the importance and use of journaling for daily life. This includes a wide range of activities from simply remembering facts to cherishing and sharing special events. Journaling can also play an important role in improving the mental health of users (Ayobi et al., 2017; Bryant et al., 2005; Isaacs et al., 2013; Lee et al., 2020; Narain et al., 2020).

Isaacs et al. (2013) introduced the term Technology-Mediated-Reflection (TMR) which highlights how technology can be used to encourage users to revisit their past events and review them. Digital journaling applications take text, voice, or image inputs and different interaction styles such as question-answering systems or chat-bots (Backer et al., 2020; Kim et al., 2019b; Lee et al., 2020; Sorapure, 2015). Some digital journaling applications (Elsden et al., 2016; Petrelli et al., 2009) have search functions combined with tagging where the entries can be categorized into

different sections such as food, people, or places. Ayobi et al. (2018), further highlighted implications for bullet journals that improve self-tracking capabilities. Our aim is to further support digital journaling applications by providing a way to create summaries of journal entries.

Another context in which summarization of autobiographical text is required is in mental health text summaries. Clinical diagnosis of mental health often relies on a patient's narrative, and the ability of a trained mental health professional to analyse and assess large amounts of such clinical data. To reduce this burden on mental health professionals, developing techniques to summarize autobiographical narrations of patients, has emerged as a need (Manas et al., 2021; Sotudeh et al., 2022). With the increasing popularity of online platforms, many use platforms such as Reachout and Reddit to share their mental health conditions, express their feelings, and seek help from the community and counselors. Reddit provides subreddit threads where the users publicly but anonymously post their mental health distress. A concise and informative summary of such posts is needed for easier and faster processing by the counselors (Sotudeh et al., 2022). Through this work, we investigate models to automatically generate summaries of such autobiographical posts by testing our methods on such a mental health post summarization dataset (Sotudeh et al., 2022).

2.2. Automatic Text Summarization

Automatic Text Summarization (ATS) (Lewis et al., 2019; Zhang et al., 2020) is a challenging research area in Natural Language Processing (NLP) (Brown et al., 2020) that mainly studies how to represent long text descriptions in compressed form without losing important information. Automatic summarization can be carried out in two ways, extractive summarization (Liu, 2019) and abstractive summarization (Zhang et al., 2020). Extractive methods attempt to summarize articles by selecting a subset of words or phrases from the original texts that retain the essential facts, while abstractive methods generate summaries based on semantic understanding, sometimes using words that were not present in the source texts. Although abstractive summarization is generally challenging compared to extractive methods, recent work has highlighted how abstractive methods can create more human-like outputs, and argue that humans prefer abstractive summaries (Keskar et al., 2019). Inspired by this, we focus on abstractive summarization in this work.

2.2.1. Self-Supervised Learning (SSL) for abstractive text summarization:

Development of Self-Supervised Learning (SSL) (Devlin et al., 2019; Kolesnikov et al., 2019; Völske et al., 2017) techniques have elevated the performance of abstractive summarization techniques significantly (Liu, 2019). SSL algorithms have two stages, pre-training and fine-tuning. The pre-training stage utilizes a selected set of pretext tasks and a large quantity of unlabelled data. Such pretext tasks

use regularities and connections in existing data to design a supervisory signal. Tasks like finding masked words in a sentence (Devlin et al., 2019) or generating the original sentence after deletion of part of the sentence (Lewis et al., 2019) can be stated as examples for pretext tasks used in the NLP domain. After pre-training, the second stage involves fine-tuning pre-trained SSL models for downstream tasks that have limited numbers of labeled examples. Recent literature describes how massive SSL models that consist of billions of parameters like GPT-3 (Brown et al., 2020) outperform baseline models in different generative NLP tasks. Although training such models from scratch is a computationally expensive and time-consuming task (Narayanan et al., 2021), most pre-trained SSL models are made available to the research community by leading research institutes. A recent work (Gururangan et al., 2020) also highlighted the influence of task-specific datasets and fine-tuning methods when using large language models to solve different downstream tasks. Our work is based on a pre-trained SSL language model called BART that has shown excellent capabilities in text generation tasks such as summarization, conversation generation, and question answering (Lewis et al., 2019; Raffel et al., 2020; Zhang et al., 2020).

2.2.2. Usage of BART model in summarization

BART is a Transformer-based (Vaswani et al., 2017) sequence-to-sequence SSL model that has shown promising results in NLP benchmarks (Wang et al., 2018) including both classification and generative tasks. The BART model has been trained with 160GB of unlabelled text data with several pretext tasks that involve corrupting text with an arbitrary noising function and learning to generate the original text from the corrupted text. BART (Lewis et al., 2019) authors have evaluated the abstractive summarization capabilities of the model for news text summary, on two benchmark news summarization datasets: XSum (Narayan et al., 2018) and CNN-DM (Hermann et al., 2015). A recent work (Cachola et al., 2020) has also shown how the BART model can be fine-tuned with a smaller amount of annotated scientific papers to generate TL;DR (Too Long; Didn't Read) summaries. These prior works have motivated us to employ BART pre-trained model in this work, to generate autobiographical summaries, but through a different strategy of fine-tuning.

2.3. Abstractive summarization datasets

There are several types of abstractive summarization datasets that belong to different domains, such as medicine (Esteva et al., 2020), scientific papers (Cachola et al., 2020), legislation texts (Eidelman, 2019), and news articles (Hermann et al., 2015; Narayan et al., 2018). Existing work (Lewis et al., 2019; Raffel et al., 2020) in ATS have mainly used two news summarization datasets (Hermann et al., 2015; Narayan et al., 2018), XSum (Narayan et al., 2018) and CNN/DM (Hermann et al., 2015) as the benchmark datasets because they contain higher quality summaries, compared to

other datasets. Although most summarization datasets are based on structured text descriptions, there are datasets made out of unstructured texts from Reddit posts (Kim et al., 2019a; Völske et al., 2017) and YouTube Video transcriptions (Sanabria et al., 2018). Due to practical difficulties of getting expert-written summaries, most of these datasets are created using alternative information; for example, using titles and post descriptions as summaries. One such dataset is Mental Health Summarization dataset, MentSum (Sotudeh et al., 2022) containing over 24 k carefully selected user posts from Reddit, along with their short user-written summary (called TLDR) in English from 43 mental health subreddits. In this work, we collect a larger amount of data (300 k posts) scraped from Reddit posts along with their titles that belong to five special sub-Reddit threads related to mental health. We also use the CNN/DM (Hermann et al., 2015) and XSUM datasets (Narayan et al., 2018) as anchor datasets due to their high-quality annotations. Details about the datasets used in our experiments are provided in Section 3.1.

2.4. Evaluation of summarization models

The most common automatic evaluation method for abstractive summarization algorithms is the Recall-Oriented Understudy for Gisting Evaluation (ROUGE) score (Lin, 2004; Lin & Ng, 2019). It consists of several metrics that are based on the lexical overlap between generated summaries and reference summaries. The most commonly used ROUGE metrics are ROUGE-1, ROUGE-2 and ROUGE-L. ROUGE-1 and ROUGE-2 refer to the overlap of uni-grams and bi-grams between the system summary and reference summary, respectively. ROUGE-L measures the longest matching sequence of words between the system summary and reference summary using the Longest Common Subsequence (LCS) statistics.

One of the most challenging area in abstractive summarization is factuality and faithfulness (Cao et al., 2018; Durmus et al., 2020; Kryściński et al., 2019). Recent works (Maynez et al., 2020; Radford et al., 2019) have highlighted how SSL-based large generative language models can generate content that is not present in the source text. These are known as “hallucinations,” which can totally change the meaning of the source text. There are also recent works that focus on evaluating the faithfulness and factuality by adapting automatic metrics like question and answering systems (Durmus et al., 2020; Maynez et al., 2020; Wang et al., 2020). They highlight how factual consistency and faithfulness depend on the summary quality of the summarization dataset used in the training process (Durmus et al., 2020) and how factual consistency and faithfulness are higher when using high-quality news summarization datasets (Kryściński et al., 2020).

Another metric for evaluating the quality of the generated text is BARTScore (Yuan et al., 2021) that uses an encoder-decoder based pretrained language model (BART) to re-score the generated text, and compares the re-scored text with the ground truth text. They argue for a formulation of

evaluation of generated text as a text generation problem by directly evaluating text through the lens of its probability of being generated from or generating other textual inputs and outputs. This is an unsupervised method to evaluate the text from different perspectives.

However, the applicability and accuracy of such automatic evaluation metrics depends on the ground-truth summaries being correct. In the Reddit post-based datasets, that are autobiographical and that will be used in this work, the “ground-truth summaries” are extracted from TL;DRs (Sotudeh et al., 2022), or post titles (Kim et al., 2019a). As it is not required of the users of Reddit to provide gold-standard perfect summaries of their posts in the TL;DRs and titles, therefore these ground-truth summaries are weak substitutes for the actual summaries. These may not always be perfect as the users may include additional information to the TLDRs and titles, other than summarizing the information in the post. This results in the ground-truth summary containing hallucinations, thus making them imperfect. Therefore, in this work, although we use ROUGE scores to get an approximate estimate of the summary quality, we heavily rely on human subjective evaluations of the generated summaries because human feedback can provide insights into the quality of the generated summaries without having to rely on the imperfect ground-truth summaries.

We reviewed various works in the literature that assess the quality of summaries through human evaluation studies. The recent automated abstractive text summarization works (Kryściński et al., 2018; Paulus et al., 2018; Tan et al., 2017) have used a combination of four criteria, namely *informative*, *concise*, *coherence*, and *readability*, with one question per criteria, such as “how informative is the summary?” (for the criteria *informative*), each to be scored between 1 and 5 or 10. However, such scoring methods suffer from a lack of a clear meaning of the different scores. Earlier literature from behavioral sciences and psychology (Friend, 2001; Garner & McCaleb, 1985; Winograd, 1984) used a similar set of criteria, however with descriptive labels for the scores to grade handwritten summaries. These descriptive labels were adopted in a three-point summary grading rubric in the more recent literature on evaluating crowd-sourced summaries (Li et al., 2018). In this work, we applied the criteria, questions, and grading rubric designed by the Document Understanding Conference (DUC) text summarization forum (Dang, 2005; Bommasani & Cardie, 2020) which has been the forum for researchers in text summarization to compare methods and results (Iskender et al., 2021).

3. Methodology

In this section, we present two approaches to generating summaries of autobiographical texts. In one approach, we use the short titles of autobiographical text (Reddit-Title) to learn the characteristics of autobiographical summaries. In the second approach, we use grammatically correct, factual, and comprehensive gold-standard summaries of the structured news text (CNN/DM (Hermann et al., 2015) along with the short titles of autobiographical text (Reddit-Title),

so that the model is able to learn the properties of both of these pairs, i.e., it should be able to generate short, usable and accurate summaries for autobiographical text while preserving the fundamental properties of abstractive summarization learned from the news-summary dataset.

3.1. Datasets

Here we discuss the datasets that we use for training (i.e., fine-tuning the BART model) and testing phases. A summary of these datasets is provided in Table 1. For different experiments, we train the model with similar amounts of data so as to avoid data bias.

3.1.1. Training data 1: Data collection from sub-Reddit threads close to autobiographical text

None of the existing large-scale datasets are created specifically to summarize autobiographical texts. For example, the Reddit TLDR (Völske et al., 2017) dataset consists of a wide range of topics including technology, gaming, finance, and well-being, and contain many non-autobiographical data points. Reddit-TIFU (Kim et al., 2019a) dataset on the other hand is limited to a single subreddit thread and is small in size. Ouyang et al (Ouyang et al., 2017), introduced a dataset with human-annotated narrative text summaries that consist of only 500 examples. Therefore, we decided to build a dataset that contains autobiographical text entries. We followed prior work (Balani & De Choudhury, 2015; De Choudhury & De, 2014; Kim et al., 2019a; Völske et al., 2017; Yang et al., 2019) that highlight the genuineness, emotionality, self-descriptive nature and the sentimentality of the stories available in some specific sub-Reddit threads. We used the sub-Reddit finder tool³ to select five publicly available sub-Reddit threads⁴, namely, “CasualConversation,” “confessions,” “mentalhealth,” “offmychest,” and “self.” We selected these sub-reddit threads after carefully considering each of their content, specifically data from these five sub-Reddit threads provided a more general and diverse dataset with entries written in autobiographical context by people all around the world. The writing style of these posts are closer to the linguistic structure of journal entries than other existing datasets. Each Reddit post we collected is tagged with a title written by the author. We refer to this collection of data as the *Reddit-Title* dataset. In total, we downloaded over 500,000 posts. We applied some data pre-processing on these posts and removed those that were empty, or only consisted of links, or had titles that are longer than their respective entries, or consisted of less than 50 words and had titles shorter than 10 words. This gave us a cleaner final dataset consisting of 263,225 entries.

3.1.2. Training data 2: Use of CNN/DM news dataset

Training an abstractive summarization model without Gold-Standard summaries (expert-written) is a very challenging task, and such models can end up generating factually inconsistent and untrustworthy summaries (Durmus et al., 2020; Lin & Ng, 2019; Maynez et al., 2020; Wang et al.,

2020). For example, the title or TL;DR of Reddit posts can serve as a weak substitute for a summary of the post. Therefore, fine-tuning the BART model solely on Reddit entries without gold standard summaries may not always be reliable. Therefore, in one of our approaches, we use a high quality CNN/DM (Hermann et al., 2015) news dataset that consists of 275,000 news articles along with expert-written summaries as an anchor in our training mechanism. This was motivated by the recent work (Durmus et al., 2020; Wang et al., 2020) that highlights how abstractive summarization models trained on CNN/DM news dataset tend to generate more trustworthy and grammatically consistent outputs compared to other datasets.

3.1.3. Test data: MentSum dataset

We used publicly available MentSum (Sotudeh et al., 2022) dataset as our unseen test dataset. This dataset is created from a set of special sub-Reddit threads that are related to mental health, that are expected to have first-person accounts of events and incidents. During the creation of the dataset, authors only selected reddit threads with a TL;DR (Too Long; Didn't Read.) post after the entry. The authors used the TL;DR as the summary. However, these TL;DRs may not always provide perfect summaries as humans often tend to include additional information to the TL;DRs other than summarizing the information in the post. Since this is the closest available dataset to a human-annotated summary of first-person autobiographical text, therefore we use this as the test set in this work. Also, we rely more on human evaluation to assess the quality of the machine-generated summaries, rather than comparing them with the less reliable ground-truth summaries of this dataset.

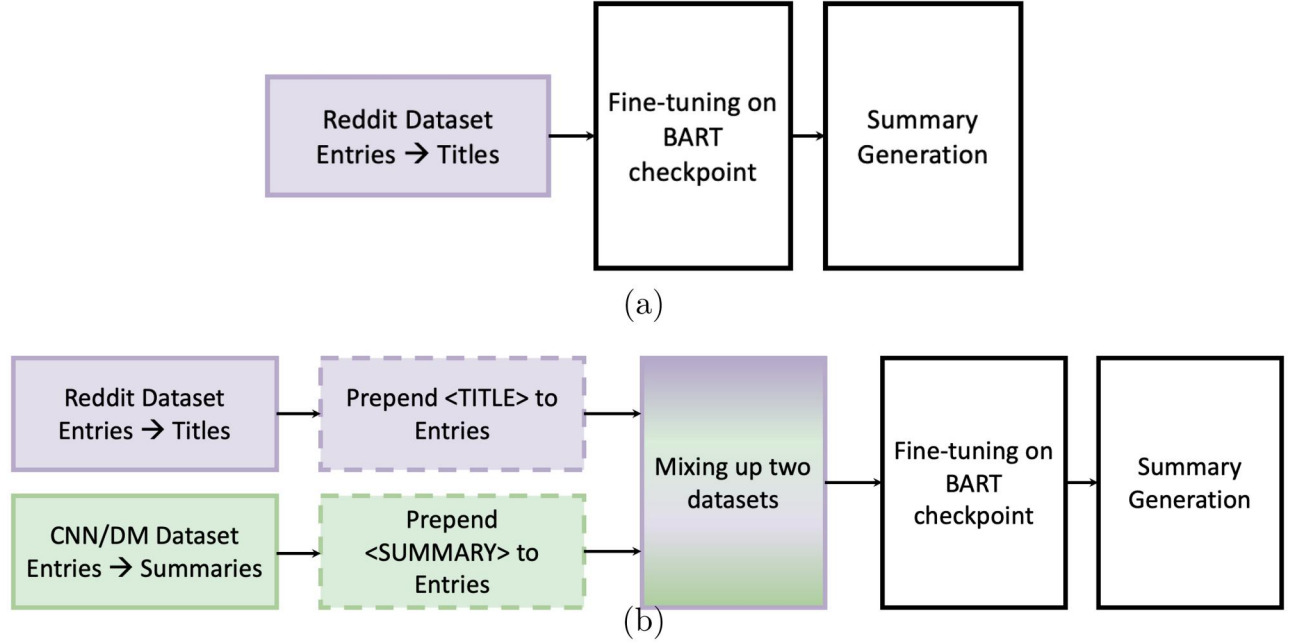
3.2. Models

We select general-purpose SSL, BART model (Vaswani et al., 2017), as our base model. In the first approach, we adapt the BART model, to the domain of autobiographical text by fine-tuning it with the autobiographical data, i.e., Reddit-title dataset (Section 3.1.1), as shown in Figure 1(a).

In the second approach, we use a mix distribution of unstructured autobiographical text and structured news text, as shown in Figure 1(b). This fine-tuning strategy is motivated by the recent work of Cachola et al. (2020) on extreme summarization of scientific documents. Cachola et al. used input data distribution from the content of a research paper and trained a model to generate the respective titles and annotated summaries. In contrast, we used two input data distributions, one, from news articles that consist of high-quality summaries (CNN/DM dataset—Section 3.1.2), and the second, from the scraped Reddit posts containing autobiography-like entries with corresponding titles (Reddit-Title dataset—Section 3.1.1). Prior to the mixing of the two datasets, we prepended two special tokens `< TITLE >` and `< SUMMARY >` to the source data stemming from Reddit-Title and CNN/DM respectively (see Figure 1(b)). The intuition behind adding this token is to have a dataset-specific

Table 1. Details about the datasets used in our experiments.

Dataset	Purpose	Content type	Number of examples	Validation split
1. CNN/DM + XSUM	Train/Fine-tune	News	300 K	20%
2. Reddit-Title	Train/Fine-tune	Autobiographic	300 K	20%
3. Reddit-Title + CNN/DM	Train/Fine-tune	Autobiographic + News	300 K	20%
4. MentSum	Test	Autobiographic	24 K	–

**Figure 1.** Overview of the (a) fine-tuning with autobiographical text dataset (Reddit-Title)—autobiographical-only model, (b) mixed distribution multitask learning method—mix model.

constant at the same location of each input that will help the model learn to distinguish data points from the two data distributions. Such a tokenized conditioning method has been used in some recent works (Brown et al., 2020; Cachola et al., 2020; Keskar et al., 2019; Raffel et al., 2020), and we have similarly used that mechanism to distinguish between data distributions. The process of conditioning input data with tokens in the training phase allowed us to generate conditional outputs in the inference phase. Therefore, tagging an input with *< SUMMARY >* at the inference stage will prompt the model to generate expert-written-type summaries.

3.3. Implementation details

Our pipeline was implemented using the Pytorch and Fairseq toolkit (Ott et al., 2019), and we chose the same hyperparameters used by the Fairseq BART implementation⁵ to fine-tune BART with the CNN-DM dataset. The BART model architecture consists of 406 million parameters, and we used a Label Smooth Cross Entropy (Müller et al., 2019; Pereyra et al., 2017) loss function of the BART model with the smoothing parameter of 0.1 similar to the BART (Lewis et al., 2019) paper. After mixing the two datasets, a total number of 550,947 data points were used for training. Since the CNN/DM dataset consists of 13,368 validation examples, we randomly selected another 13,368 examples from the

Reddit-Title dataset that are uniformly distributed among the five sub-Reddit threads to balance the validation. Therefore, our validation set had 26,736 examples in total. The final fine-tuning was done with 8 parallel V100 32GB GPUs and completed in 8 h. Since our combined dataset is twice the size of CNN/DM, we changed the number of warm-up updates from 0 to 1000, and total updates from 20,000 to 40,000. Our final checkpoint, along with the entire source code is Open-sourced⁶ for the public.

4. Evaluation

4.1. Technical evaluation

We conduct the technical evaluations using ROUGE scores (Lin, 2004) and BART scores (Yuan et al., 2021) to address the question of whether domain-adapted models outperform a baseline model trained only on structured news data. However, we acknowledge that the ground-truth summaries in the test dataset (Section 2.4) would be imperfect as they may have hallucinations, in the same way as other widely used Reddit datasets, such as Reddit-TIFU (Kim et al., 2019a) have, therefore we rely on human evaluations to confirm the quality of the summaries generated from the domain-adapted models.

Details of the datasets used for our experiments are given in Table 1. We make sure that the number of training examples remains the same across different models.

Table 2. Performance of systems with different fine-tuned BART model variations in terms of ROUGE and BART scores (higher the better).

Model	BART Fine-Tuning Variations	Rouge-1	Rouge-2	Rouge-L	BART score
1	Only on news (CNN/DM + XSUM)	21.6	3.7	18.4	−3.96
2	Only on Reddit-Title (autobiographical-only model)	24.0 [†]	4.9 [†]	19.6 [†]	−3.44 [†]
3	On a mix of CNN/DM and Reddit-Title (mix model)	23.6 [†]	4.9 [†]	19.6 [†]	−3.48 [†]

The test dataset is 10% of the MentSum (Sotudeh et al., 2022) dataset (1636 samples). [†]Denotes statistical significance ($p < 0.05$) with respect to model 1.

4.1.1. Metrics

ROUGE score is the most common text-summarization evaluation technique in Natural Language Processing (NLP) (Lewis et al., 2019) that calculates the similarity or the lexical overlap between a candidate text and a reference or ground-truth text. We used three ROUGE score metrics (Lin, 2004) for evaluation, which measures the alignment of uni-grams (Rouge-1), bi-grams (Rouge-2) and n-grams (Rouge-L) between generated text and human-written text. Each Rouge score calculation gives precision, recall, and F -scores. We report the F -scores of these metrics in Table 2 for four different models where we compare the model-generated summaries to the ground-truth summary of the MentSum dataset. Higher the score, the more the similarity between the generated text and the human-written text.

We compute BART scores (Yuan et al., 2021)⁷ as (equally) weighted log probability of one text y given another text x . Specifically,

$$\text{BARTScore} = \sum_{t=1}^m \omega_t \log p(y_t | y_{<t}, x, \theta) \quad (1)$$

Where, given a seq2seq model parameterized by θ , source sequence containing n tokens $x = x_1, \dots, x_n$ and target sequence containing m tokens $y = y_1, \dots, y_m$. Since the BART Score is the average log-likelihood for target tokens, the calculated scores will be smaller than 0 (the probability is between 0 and 1, so the log of it should be negative). The higher the log-likelihood, the higher the probability, and the better the summary. Due to its generation task-based formulation and use of pre-trained BART model, BARTScore has been shown to be flexibly used in different evaluation scenarios based on different generation directions, such as Precision (measures how likely the system-generated summary could be constructed based on the ground-truth summary), Recall (quantifies how easily a ground-truth summary could be generated by the system-generated summary), and F -Score. We report the F -Score which uses the arithmetic average of Precision and Recall and evaluates the semantic overlap (informativeness, adequacy etc.) between the ground-truth and generated summary texts.

4.1.2. Results

First is the baseline model which is the BART model fine-tuned on news data selected from CNN/DM and XSUM datasets (Table 1 dataset 1). The second model is domain-adapted where the same BART model is fine-tuned on only

Reddit-Title data (Table 1 dataset 2), called autobiographical-only model. The final model was trained with a mix of Reddit-title data and CNN/DM (Table 1 dataset 3), called mix model. The ROUGE and BART scores are presented in Table 2. Through both the ROUGE and the BART scores, we found that the domain-adapted models (Table 2 models 2 and 3) perform significantly better ($p < 0.05$) than the model fine-tuned only on news data (Table 2 model 1) (pair-wise significance t -tests). This outcome was expected considering the similarities in data distributions between the Reddit-Title training data and the MentSum test dataset. However, both the kinds of domain-adapted models—autobiographical-only and mix models show similar performance for both ROUGE and BART scores, and the difference between them is not statistically significant ($p > 0.05$).

These technical evaluation metrics rely on accurate ground-truth summaries, however, Reddit-based datasets are not manually curated to guarantee accurate ground-truth summaries. Therefore, we cannot entirely rely on the Rouge scores computed on these summaries to assess their quality. Thus, we conducted human evaluation experiments to assess the quality of the generated summaries, as explained in the next subsection.

4.2. Human evaluation

Since the technical evaluations can be considered to be ‘weak’ because of the unreliable ground-truth summaries, we conducted a comprehensive evaluation study with human participants as our main method of assessing the quality of the generated summaries, independent of the ground-truth summaries. We used Amazon Mechanical Turk,⁸ a crowdsourcing platform, to recruit participants as well as conduct the evaluation study online.

4.2.1. Experimental setup

In our MTurk study, each Human Intelligence Task (HIT) involved reading the source text first and then evaluating the two summaries (one generated by the mix model, and the other generated from the autobiographical-only model) presented together on the basis of a set of 12 questions. The workers on MTurk did the task remotely on their own computers while interacting with our hosted HIT as web content via the Amazon MTurk website.

4.2.1.1. Data. We compared summaries generated from two of the domain-adapted models—model 2, i.e., *autobiographical-only*, which is a transfer-learning-based model fine-tuned only on Reddit-Title data, and model 3, i.e., *mix*, which uses the mix-distribution multi-tasking method. In this experiment, we first randomly selected 50 source texts from the MentSum test dataset—which is an unseen dataset for both models—and generated two summaries for each source text from the two summarization models (A sample set of summaries is shown in Appendix Figure A1).

Table 3. Questionnaire for human evaluation studies.

Criteria	Question
Grammaticality	1. Are there any obviously ungrammatical sentences, e.g., missing components, unrelated fragments or any other grammar-related problem that makes the text difficult to read?
Non-redundancy	2. Are there any datelines, system-internal formatting or capitalization errors that can make the reading of the summary difficult? 3. To what degree does the summary say the same thing over again? Unnecessary repetition might take the form of whole sentences that are repeated, or repeated facts, or the repeated use of a noun or noun phrase (e.g., "Bill Clinton") when a pronoun ("he") would suffice.
Overall Quality	4. If you were editing the summary to make it more concise and to the point, how much useless, confusing or repetitive text would you remove from the existing summary?
Structure and Coherence	5. Does the summary state the main idea of the source? 6. Does the summary contain the important points conveyed in the source and exclude unnecessary details?
Referential Quality	7. Does the summary build from sentence to sentence to a coherent body of information about the topic? 8. Does the summary have a focus, i.e., sentences only contain information that is related to the rest of the summary?
Comparisons	9. How much trouble did you have identifying the referents of noun phrases in this summary? Are there nouns, pronouns or personal names that are not well-specified? For example, a person is mentioned and it is not clear what his role in the story is, or any other entity that is referenced but its identity and relation with the story remains unclear
Hallucinations	10. To what degree do you think the entities (person/thing/event/place/...) were re-mentioned in an overly explicit way, so that readability was impaired? For example, a pronoun could have been used instead of a lengthy description, or a shorter description would have been more appropriate? 11. Which summary do you prefer and why?
	12. Does the summary contain any information which is not present in the source text?

4.2.1.2. Study questionnaire. We prepared this questionnaire based on the grading rubric designed by the Document Understanding Conference (DUC) text summarization forum (Dang, 2005; Bommasani & Cardie, 2020). This rubric consists of 5-point descriptive grading for text summarization using four criteria, namely *grammaticality*, *non-redundancy*, *referential clarity*, and *structure and coherence* with two questions per criteria. Additionally, we included a criterion to test the *overall quality* of the summaries (Iskender et al., 2021; Louis & Nenkova, 2013) with two questions, a direct question regarding *hallucinations* (Durmus et al., 2020), and a comparative question that asks their reason for preferring one summary over the other. The list of criteria and the list of 12 questions are summarized in Table 3. Questions 1–10 have a 5-point descriptive rubric, provided in Appendix Table A1, which was presented as radio buttons on the HTML page in the MTurk screen (see Appendix Figure A1). Questions 11 and 12 have four options—Summary A, Summary B, Both, or None, along with a text box for question 11.

4.2.1.3. Inclusion criteria. The location for taking up the task on MTurk was set to English-speaking countries so that the probability of fluency in the English language is high.

4.2.1.4. Exclusion criteria. We added a sanity check question to the questionnaire to ensure that workers are paying attention. This question provides a sentence that is either extracted from the current source or not, and the question is whether the sentence belongs to the source text. If this question was answered incorrectly, we excluded that HIT response from our analysis. However, this did not affect the remuneration received by the workers.

4.2.1.5. Number of responses collected and remuneration. Each HIT was completed by ten workers which provided 500 evaluations altogether (i.e., 50 source-summaries $\times$ 10 workers). For each HIT, a worker was paid USD 1.0. We rejected 11 responses based on incorrect answers to the sanity check question, thus we had 489 total responses.

Table 4. Summary of human evaluation results.

Criteria	Model	Mean	Std. Dev
<i>p</i> -Value			
Grammaticality (Q1, Q2)	Autobiographical-only	4.29	0.77
	Mix	4.38	0.75
Non-Redundancy (Q3, Q4)	Autobiographical-only	4.48	0.77
	Mix	4.52	0.71
Overall Quality (Q5, Q6)	Autobiographical-only	4.04	1.04
	Mix	4.10	1.04
Structure and Coherence (Q7, Q8)	Autobiographical-only	4.48	0.76
	Mix	4.53	0.72
Referential Clarity (Q9, Q10)	Autobiographical-only	4.67	0.68
	Mix	4.73	0.61

4.2.2. Results

Table 4 shows the results of the human evaluation tests for five of the criteria. We find that both the model-generated summaries perform very well in all criteria as they are given a high average rating (> 4.0) by the human evaluators. The questions under the criteria of referential clarity were given the highest ratings (> 4.6). The questions on non-redundancy and structure and coherence got a similar range of ratings ($\sim 4.4 - 4.5$), while the questions on grammaticality got relatively lower average ratings (~ 4.3). The overall quality criteria as well as Figure 2(a) show that mix model summaries are preferred slightly more than autobiographical-only model summaries (i.e., 37% of the responses preferred the mix model, while 34% of the responses preferred the autobiographical-only model).

We conducted pair-wise significant tests between the average ratings of each criteria between autobiographical-only and mix models, and we do not find any statistically significant difference between the ratings of the two kinds of summaries, as shown in Table 4 ($p > 0.05$). However, the grammaticality criterion is borderline significant ($p = 0.08$), meaning there is a slight improvement in the grammar of the summaries generated from mix model over those from autobiographical-only model, which could be attributed to the inclusion of the structured news data in the fine-tuning process of the mix model.

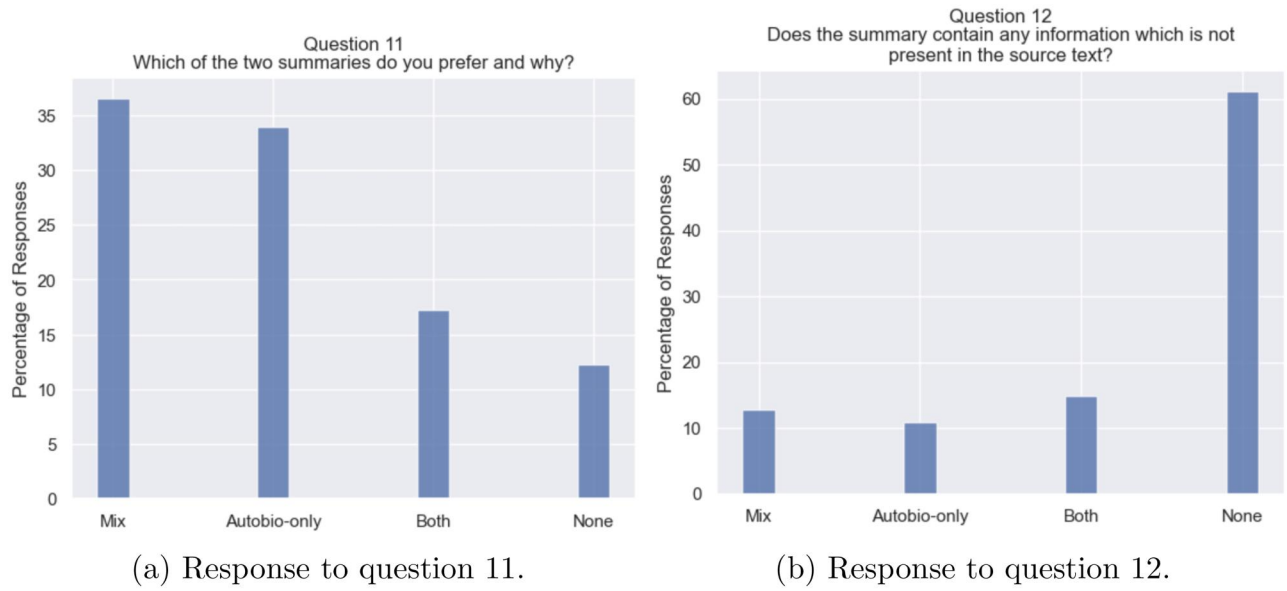


Figure 2. Human preference and perceived hallucinations for summaries generated by mix and autobiographical-only models.

Some of the answers to the question of why the evaluators preferred one summary over the other (Q11), when they chose the mix model summary as their preference, were,

Summary A did a better job of summarizing the important details in the original text with no issues with grammar.,

The words in both summaries are similar, but I prefer the order of them in Summary A.

And some answers when they preferred the autobiographical-only model summary over the mix model summary,

I slightly prefer A as it is a little more specific. B is still good though.,

B better coheres to the main point of the summary. A is a little more off track.,

Figure 2(b) shows that most of the summaries (>60% of the responses) generated by these models do not contain hallucinations. Thus, we infer that the domain-adapted BART models provide reliable summaries for autobiographical text.

5. Practical implications case study 1: A digital journaling application

To study how automatic text summarisation could impact digital journaling, we developed a prototype journaling application where we incorporated the mix model for text summarization and tested it with users.

5.1. Design and development

5.1.1. Focus group

We conducted a focus group study with six participants to understand the requirements for a proof-of-concept journaling application. We sketched a few ideas together with them to initiate a discussion. Interestingly, all the participants

preferred voice typing but acknowledged imperfections in text-to-speech systems such as errors when transcribing voice. As such, we included a voice recording button as well as a text box so that users can simultaneously voice and keyboard to enter text entries. Participants agreed that this was a reasonable way to tackle it. Four participants pointed out that seeing the summary in a new window is not ideal for them since they had to move back and forth to read the story and summary. When this concern was raised and discussed, a participant mentioned,

I would like to see a real-time summary on the side while I'm typing.

They further explained to us how it would give them feedback about their journal entry. Three other participants also liked the idea of having a real-time summary, and we decided to implement a real-time summary display function. Based on this discussion, we decided to develop a simpler UI with only two windows, one to enter the journal entry and one to review past records.

5.1.2. Development

We employed a client-server architecture to develop *SummarizeMe*. We hosted the *SummarizeMe* model as a server-side application served via the Python Flask web framework⁹ on Google Cloud. To achieve a faster response time, we used a Virtual Machine with an Nvidia P100 GPU to run the server application. The client-side was developed with HTML and JavaScript which can swiftly run on a web browser. The main UI (Figure 3) provides two major views, the first allows the user to enter a journal entry, and the second window shows all the past diary entries with the summaries. We used the Speech Recognition API¹⁰ in the web browser to implement the voice typing feature which allowed the front-end to work swiftly in multiple PC and mobile web browsers (Google Chrome, Microsoft Edge, Safari, WebView Android, Chrome Android, Safari on iOS

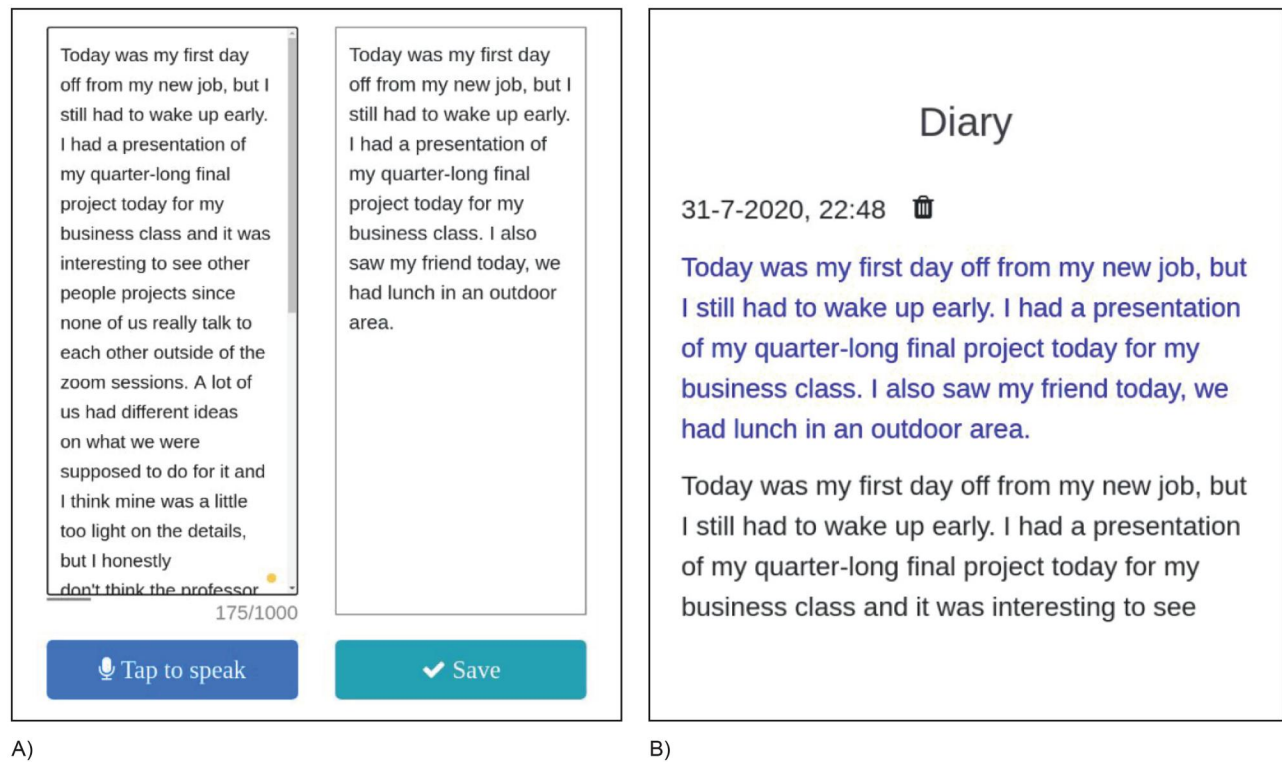


Figure 3. UI of the prototype journaling application (resized to display). (a) Journal entry field and the generated summary next to it. (b) List of saved diary entries (black text-only part of the text is shown) together with stored summaries (blue text).

and Samsung Internet). Further, to extend browser support, we used the WebKit Speech Web API.¹¹ To offer users a preview of their summarized text, we generated a summary whenever the user was idle for over 2 s and updated it when they modified the text entry. In order to secure user privacy, we only saved each user's previous entries and their summaries in the local browser using the Web Storage API.¹² This implementation allowed users to remove entries that persisted in the browser when they cleaned the cache of the web browser.

5.2. User feedback

We gave our digital application to 75 participants (40 males, 35 females). Their mean age was 28 years ($SD = 6.8$). All 75 participants rated themselves as native or fluent English speakers. Participants received a URL to our web app via email along with a video demonstrating the usage of the app. The participants were instructed to input a personal story or experience using voice or keyboard input, and then focus on the summary generated by the system. We highlighted that we did not have access to any user data to ensure their freedom to express themselves. We asked them to fill out a questionnaire to collect responses including fluency in English, the model's ability to capture the main idea, sentiments, and emotions of the story, generation of unnecessary information, and suitability for journaling application. These questions were to be answered on a 7-point likert-scale. In addition, we conducted open-ended interviews to understand their experience of our diary application. We followed the general guidelines for thematic

analysis described by Braun and Clarke (2006) to analyze the interview responses. The main findings are summarised below.

5.2.1. Users feel that the summary captures the main idea of the original text

Some answers to the second question in Figure 4 along with comments to the open-ended questions were:

It surprised me with how it combined words that I had written and how it highlighted the main points in my essay.,

It captured the essence of the story which is surprising.

These comments show that mix model captures the main idea of entries.

Interestingly, a few users had found it to work beyond expectations:

accuracy is above expectations!.

Some even thought that the summarization model works better than humans;

The application shortens the original paragraph while keeping the meaning as is. I feel like it will be a difficult task for me to beat the model in terms of summarization.

and,

I think it's super accurate. It managed to capture the key emotions that I felt. I don't think I could have summarized it as succinctly as it did. Great work!.

While some of these comments might have been influenced by experiencing AI generated summaries for the first time, these results and comments validate the expected

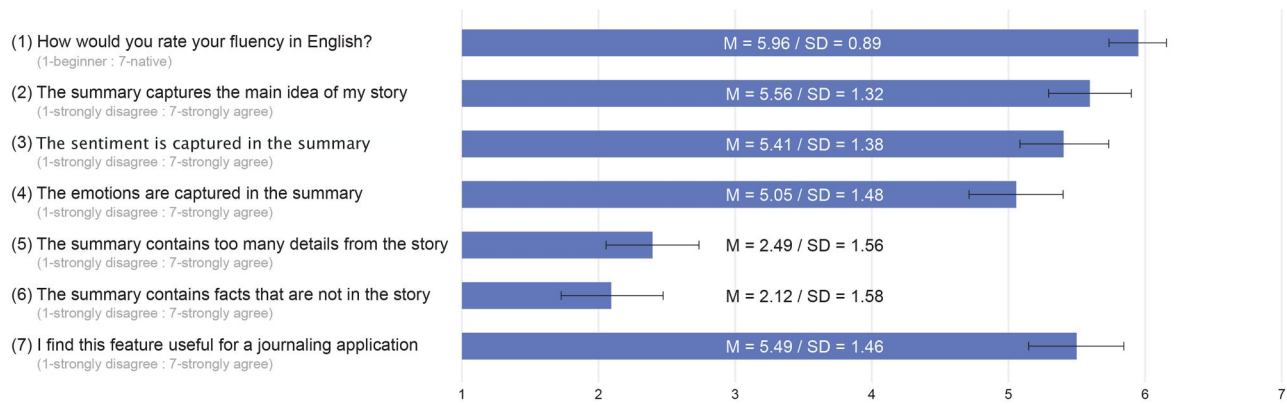


Figure 4. Responses for the 7-point likert scale questionnaire. Mean-bar plots are shown with the 95% confidence interval. Responses were collected from 75 participants.

performance of the summarization task primary task, which is capturing the essence.

5.2.2. The preferred length and amount of key details are user-dependent

As shown by answers to the question 5 and 6 in Figure 4, auto generated summaries have rarely generated new information that was not in the original entry, and not included too much details. Prior work shows that the trustworthiness of a model is essential for real-world applications (Durmus et al., 2020; Kryściński et al., 2019). We used CNN/DM dataset in the training process to achieve this, and we believe this was fulfilled. However, some users thought that some details about the key entities like names of places should be included in the summary. Some participants thought that short and concise summaries are not useful for them, and they rather preferred to have a more detailed summary. By analyzing comments regarding these possible improvements, we identified future opportunities to further fine-tune and explore ways of parameterizing the generated summaries. This will enable the users to adjust the parameters to receive desired types of summaries.

5.2.3. The summary preserves affective content

According to the answers to the question 3 and 4 in Figure 4, users indicated that the sentiment of the original text had been preserved in the summary, and emotions were also preserved to a certain extent. Some users even liked to have an analysis of their emotions as one commented,

I think, having a simple analysis about the emotions (eg: happy-60%, angry 10 sad 5% etc) out of the para [paragraph] would be a great add-on..

These results show that it is important to preserve the affective information of the original text. We also noticed a strong correlation ($r = 0.64, p < 0.01$) between the self-reported level of fluency in English and score of perceiving that the sentiments are preserved in the summary.

5.2.4. Continuous summary generation can be distracting

While many found the summaries to be good enough to be saved as is, a few participants mentioned that they sometimes edited the summary before saving. However, they still found it much easier than writing the entire summary themselves. Some participants liked the real-time summary generation feature while some disliked due to being distracted by it when typing the diary. There could be a sweet spot on generating the summary by experimenting on the idle time. Alternatively, real-time summary generation can be made optional by letting the user decide that setting.

5.2.5. Quickly glancing at summaries is helpful to reflect

Majority of the participants found that the auto summarization helps reduce the information overhead problem by providing them the necessary means to quickly glance at records and reflect. For example,

When read after a few days the summaries were able to spark reminders and helped me trace-back the thought train. This was especially helpful in me reaching a state of self-realization on certain issues I had in mind..

These suggest that auto summarization has the potential to be deployed in real-world digital journaling applications to tackle the information overload problem.

5.2.6. Real-time, summarizing feedback can feel connecting

Apart from general aspects, some feedback such as

I really felt connected with the application. The summary got the key points and suddenly I am not talking to just a machine. I am getting more expressive

showed how some participants were “connecting” with the application. One participant mentioned,

When I read the dairy, summary says it all (laughs). I thought some things are big deals, but actually it is just an incident.

This is interesting since it shows that having a summary can make people see problems from outside the box. We believe that auto summarization feature can be directly used with previous applications like “Echo” (Isaacs et al., 2013) that encourages users to reflect on their diary entries.

5.2.7. Privacy concerns are not deal-breakers

Interestingly, the majority of the users did not worry about data-privacy as they knew we did not have access to their data. However one user re-confirmed with us that the access to their stories were not available to anyone else, including the research team. Based on conversations, we also saw the potential of using their data within a secure cloud service to improve the model further in future. Many users were aware how cloud based applications worked and had knowledge about how applications collected data to improve services and push advertisements. It is really important establish user-trust and have a privacy policy that explains clearly how the data is managed.

5.2.8. Play with AI

Interaction with a generative language model is not common, and it may have been the first experience of some users to playing with AI. We came to this conclusion after noticing that some participants had experimented on the summarization model. For example, one participant mentioned,

I input previous entry and added some excess, meaningless words to the entry. It is amazing to see how the algorithm drop those meaningless information and gave me the same output.

While some users were amazed by the level of performance, some users desired more functionalities. This shows how different people expect automated systems to perform at different levels. This can be due to prior experience with similar technologies, and how much people take systems for granted. This is an interesting and important consideration when talking about user adaptation of AI products or services.

6. Practical implications case study 2: Comparison with ChatGPT

Recent studies have assessed the performance of ChatGPT, a commercial conversational AI system, on various factual datasets on particular subjects such as medicine, finance, and law (Guo et al., 2023). In this work, we empirically study the performance of the mix model and ChatGPT to assess the advantages and disadvantages of these systems in the context of autobiographical text. Due to its proprietary nature, the ChatGPT model is not comparable to the mix model due to differences in model size and training data. For example, the ChatGPT model has more than 175B parameters (Brown et al., 2020), while the BART model has about 406M parameters (Lewis et al., 2019). However, it is worthwhile to note and analyse how humans respond to the summaries generated by these models as this exercise will indicate the utility of a general-purpose chatbot in the context of summarizing autobiographical text, as well as the pros and cons of a dedicated autobiographical text summarization model.

We conducted human evaluation experiments using the same questionnaire as described in Section 4.2. We selected 35 source texts from the MentSum test dataset, and generated summaries from the mix model and ChatGPT, which we used in the questionnaire in MTurk to conduct the human evaluation study. We conducted this study in February 2023, and to generate summaries using ChatGPT, specifically, we

used its most powerful engine available at the time—“text-davinci-003”, which was the version just before GPT3.5¹³. To achieve this, we used the prompt—“[Source Text] + Summarize the above text using 33 to 48 words”. The word range 33 to 48 was determined by calculating the mean and the standard deviation of the ground truth summaries of the MentSum dataset, which were 40.32 and 6.93 respectively. The max token count was set to 1024 as the total number of words between the prompt and the generated summary is less than 750 words across all the data points. We were interested in evaluating the summaries generated by ChatGPT in its default mode, and hence we did not experiment with the prompt or fine-tune the model. We obtained 350 evaluation responses (10 workers per task), where we applied the sanity check criteria and obtain 344 responses to the questions.

Table 5 shows the results of the human evaluation tests for five of the criteria. We find that for four of the criteria, ChatGPT summaries significantly outperform mix model summaries. This is also reflected in the response to Q11 (Figure 5(a)) where ChatGPT summaries are preferred over mix model summaries. However, we get a high average rating of the mix model summaries from the human evaluators for most of the criteria (>3.9), even when the mix-model summaries are compared against the summaries generated from ChatGPT. From the responses to Q12 in Figure 5(b), we find that about 10% of the summaries generated from ChatGPT also show hallucinations. These observations imply that a relatively small-scale and open-source model that is tailored for the purpose of summarizing autobiographical text gives convincing summaries that are comparable in quality to that generated by a commercial large-scale and closed-source general-purpose AI.

Some of the answers to the question of why the evaluators preferred one summary over the other (Q11), when they chose the mix model summary as their preferred summary over the ChatGPT generated summary, were,

Summary A was more organized and easy to understand the point.,

It is in the first person. This makes it compelling.,

Summary A states the more personal elements of the source text. It's more interesting.,

The first-person format is more friendly and readable.,

Summary A asks a question at the end that is not in the original text. Summary B is much better..

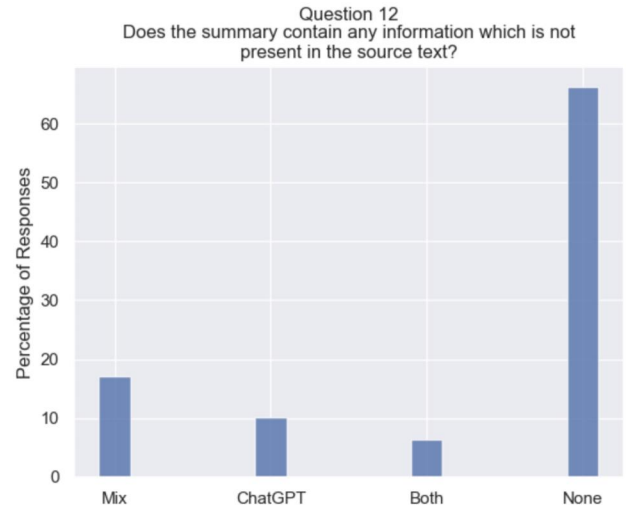
And some answers when they preferred the ChatGPT summary over the mix model summary,

Table 5. Human evaluation MTurk experiments of mix Vs ChatGPT.

Criteria	Model	Mean	Std. Dev	p-Value
Grammaticality (Q1, Q2)	ChatGPT	4.34	0.68	0.04
	mix	4.22	0.77	
Non-Redundancy (Q3, Q4)	ChatGPT	4.62	0.63	0.02
	mix	4.50	0.73	
Overall Quality (Q5, Q6)	ChatGPT	4.27	0.89	0.00
	mix	3.91	0.93	
Structure and Coherence (Q7, Q8)	ChatGPT	4.63	0.64	0.00
	mix	4.50	0.73	
Referential Clarity (Q9, Q10)	ChatGPT	4.72	0.69	0.25
	mix	4.66	0.77	



(a) Response to question 11.



(b) Response to question 12.

Figure 5. Human preference and perceived hallucinations for summaries generated by mix and ChatGPT models.

I think this summary is the one that sticks to the source unlike summary B.,

While the writing style in A is a bit clunky, it's just much better than B overall because B leaves out way too much to be a good summary of the source, and also has that x-post stuff,

Summary B is too short. Summary B gives more of the relevant details from the original text - therefore, I feel that it is the better summary.,

Both summaries are well written, but summary A mentions ADHD and summary B leaves out that detail, so I felt that summary A was better overall.,

B is slightly more coherent and does a better overall job of sticking to the original sentence..

In summary, the human evaluators liked the first-person tone of the summaries generated by the mix model, possibly because this tone matched that of the autobiographical source texts. However, ChatGPT tended to present the summary in a third-person perspective, eg. 'The writer is ...', 'The narrator was'. This highlights the need for tailoring such a general-purpose model for specific and personalized tasks. One could modify the prompt to ChatGPT to get more first-person outputs or possibly fine-tune ChatGPT with autobiographical text. A clear advantage that mix model provides in such a scenario is that it is open-source that facilitates easier and cost-efficient ways of fine-tuning for a specific and personalized task. Moreover, there are privacy and data concerns when employing such commercial general-purpose models running on the cloud for sensitive personal data such as journal entries.

7. Discussion

7.1. Contributions

In summary, our contributions are:

- Extending an open source general-purpose generative language model to work with "autobiographical text" by incorporating a weak supervised signal, for which:

- We build a new autobiographical text dataset—*Reddit-Title* (web-scraped and cleaned Reddit dataset with 300,000 data points) for the research community, which will make research for autobiographical text easily accessible and reproducible
 - We assessed and compared the performance of two models of domain-specific fine-tuning for autobiographical text summarization, through technical metrics as well as human evaluations. We found that both of these models which were adapted using our autobiographical text dataset outperformed the model adapted with the existing structured news datasets.
- Investigating the practical implications of our approach of autobiographical text summarization, for which,
 - We incorporated our autobiographical text summarization model in a proof-of-concept digital journaling web application and studied its implications in the context of individual digital journaling.
 - We presented a comparison of the summaries generated from this fine-tuning strategy with those from a commercial product (ChatGPT) through human evaluation tests.

7.2. Limitations and future work

We note that there are many other advanced techniques such as PEGASUS (Zhang et al., 2020), ProphetNet (Qi et al., 2020), and T5 (Raffel et al., 2020). However, our aim was to show the value of our dataset in fine-tuning a general-purpose SSL.

Further improvements are necessary for the mix model because the Reddit data that it is fine-tuned on is noisy despite the data pre-processing steps, which results in noisy and unrelated words in the generated summaries, as also noted by the human evaluators in the MTurk experiments. Moreover, the ground-truth summaries of the Reddit-title

dataset that were used for fine-tuning could contain hallucinations resulting in hallucinations in the generated summaries from the mix model.

Furthermore, the lack of autobiographical datasets with gold-standard summaries limits the possibility of evaluation of the summaries via comparative technical metrics such as ROUGE scores or more sophisticated metrics such as BARTscores (Yuan et al., 2021). Future work would include building a curated dataset for autobiographical text summarization that will help in benchmarking the performance of these models through technical metrics.

We developed our own proof-of-concept application to test the mix model approach to summarize autobiographical text. Our digital journaling prototype was affected by the imperfections of the speech-to-text APIs. We used the built-in speech-to-text API of the Google chrome browser. This presented several issues, such as difficulties handling various accents and the absence of appropriate punctuation. Improvement of speech-to-text technologies will further elevate the performance of our journaling application. We open-sourced our model to enable easy integration into existing applications with existing user bases.

Usually, SSL language models are large in size and require high computational power. Therefore, cloud-based approaches seem feasible rather than running these models on the device. Improvements in models like the mix model will require access to more real-world data, posing privacy challenges. However, novel techniques like federated learning (Konečný et al., 2016; Yang et al., 2019) can be leveraged to deploy with cloud services without sacrificing user privacy.

With further improvements in GPT technology (GPT4) and new initiatives such as IOS 17 Journal app, the scope of journaling has broadened to non-text data (images, photos, videos, songs etc), that needs further systematic studies for abstractive summarization capabilities.

Although our study design included randomized sampling of participants for the user studies, we had some control over the demographics of the participants (English-speaking countries in MTurk studies, and University of Auckland in Journaling Application case study) and there was a possibility of re-sampling in the different MTurk studies. Therefore, future work would include improved randomized sampling strategies with a larger number of participants to further support the generalizability and statistical validity of the results.

8. Conclusion

In this paper, we investigated several abstractive text summarization techniques for the purpose of summarizing autobiographical text such as journal entries. We study two approaches to adapt an existing SSL-based language model (BART) to summarize autobiographical text with weakly-supervised data—one, where we fine-tune the BART model with only autobiographical text that we extracted and curated from specific Reddit posts, while in the other, we mix

high-quality news summarization data with the same autobiographical text. With technical and human evaluations, we find that the summaries from both of these models have a high quality and are comparable. We incorporate the mix model in our proof-of-concept journaling application and showed that our approaches could potentially solve the information overhead problem for such applications. Moreover, we compare the quality of the summaries generated from a general-purpose commercial application ChatGPT and mix model through human evaluation tests providing insights into the practical implications of these models. We have open-sourced the mix model as well as our curated Reddit-title dataset that will allow other researchers to build upon or use it as is.

Notes

1. <https://openai.com/blog/chatgpt>
2. <https://drive.google.com/drive/folders/1T4dk3FZxlsArICLL8dRB5vcycL5KkhWF?usp=sharing>
Google Drive link for review
3. <https://valohai.com/subreddit-finder/>
4. [reddit_train_links](#).
5. <https://github.com/pytorch/fairseq/blob/master/examples/bart/README.summarization.md#fine-tuning-bart-on-cnn-dailymail-summarization-task> Link to fine tuning BART on CNN-Dailymail dataset.
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Appendix. Detailed questionnaire for human evaluation studies and MTurk study sample page**Table A1.** Questionnaire for human evaluation studies.

Criteria	Question	Answer Options
Grammaticality	1. Are there any obviously ungrammatical sentences, e.g., missing components, unrelated fragments or any other grammar-related problem that makes the text difficult to read?	1—Too many problems, the summary is impossible to read 2—A fair amount of grammatical errors 3—Some problems, but overall acceptable 4—Minor grammar problems 5—No noticeable grammatical problems
	2. Are there any datelines, system-internal formatting or capitalization errors that can make the reading of the summary difficult?	1—Many, to an extent that reading is difficult 2—A fair amount of formatting problems 3—Some, but they do not create any major difficulties 4—Minor formatting problems 5—No noticeable formatting problems
	3. To what degree does the summary say the same thing over again? Unnecessary repetition might take the form of whole sentences that are repeated, or repeated facts, or the repeated use of a noun or noun phrase (e.g., “Bill Clinton”) when a pronoun (“he”) would suffice.	1—Quite a lot; most sentences are repetitive 2—More than half of the text is repetitive 3—Some repetition 4—Minor repetitions 5—None; the summary has no repeated information
Non-redundancy	4. If you were editing the summary to make it more concise and to the point, how much useless, confusing or repetitive text would you remove from the existing summary?	1—Most of the text 2—A lot 3—Some 4—A little 5—None
	5. Does the summary state the main idea of the source?	1—The summary does not state the main idea 2—The summary touches upon a part of the main idea 3—The summary partially states the main idea 4—The summary almost completely states the main idea 5—The summary clearly states the main idea
Overall Quality	6. Does the summary contain the important points conveyed in the source and exclude unnecessary details?	1—No pieces of major supporting information stated and not necessarily in a logical order. Many minor or unimportant details or reflections. 2—Very few major supporting information stated. A number of unimportant details mentioned. 3—Some but not all major supporting information stated and not necessarily in a logical order. Some minor or unimportant details or reflections. 4—Most of the major supporting information mentioned in somewhat logical order. Almost no unimportant details mentioned. 5—Major supporting information stated economically and arranged in a logical order. No minor or unimportant details or reflections.
	7. Does the summary build from sentence to sentence to a coherent body of information about the topic?	1—Incoherent 2—Not so coherently 3—Neutral as to coherence 4—Somewhat coherently 5—Very coherently
Structure and Coherence	8. Does the summary have a focus, i.e., sentences only contain information that is related to the rest of the summary?	1—No, the summary has a bunch of unrelated sentences 2—The sentences in the summary are only related minorly 3—The sentences in the summary are somewhat related 4—The summary has mostly related statements 5—The summary is focussed and contains well-related statements
	9. How much trouble did you have identifying the referents of noun phrases in this summary? Are there nouns, pronouns or personal names that are not well-specified? For example, a person is mentioned and it is not clear what his role in the story is, or any other entity that is referenced but its identity and relation with the story remains unclear	1—Severe problems; main events, characters or places are not well-specified and/or it's difficult to say how they relate to the topic

(continued)

Table A1. Continued.

Criteria	Question	Answer Options
Referential Quality	10. To what degree do you think the entities (person/thing/event/place/...) were re-mentioned in an overly explicit way, so that readability was impaired? For example, a pronoun could have been used instead of a lengthy description, or a shorter description would have been more appropriate?	2—Rather problematic; enough events/things/ people/places are unclear that parts of the summary are hard to understand 3—Somewhat problematic; some minor events/ things/people/ places are unclear, or a very few major ones, but overall the who and what are clear. 4—Slight problems, mostly cosmetic/stylistic 5—No problems; it is clear who/what is being referred to throughout. 1—A lot: reintroduction of characters and entities made reading difficult/caused comprehension problems
		2—Rather problematic: to a degree that became distracting, entities were over-described 3—Somewhat: to a noticeable but not annoying degree, some entities were over-described 4—A little: once or twice, an entity was over-described 5—None: references to entities were acceptably explicit
Comparisons	11. Which summary do you prefer and why?	1—I prefer Summary A 2—I prefer Summary B 3—I prefer both 4—I prefer none Why?—[Textbox] 1—Only Summary A
Hallucinations	12. Does the summary contain any information which is not present in the source text?	1—Only Summary A 2—Only Summary B 3—Both the summaries 4—None of them

Source	
A close friend of mine says she will commit a suicide if she'll get kicked out of the University and her academic condition is pretty bad.. so she'll probably get kicked.... she says it not as a threat or an idea, but as a known fact that cannot be changed.. An amazingly close friend of us shot himself two months ago After he told her he thinks that committing a suicide is a stupid idea so now she thinks that's a great solution for a fucked up life.. She's 21, but she has a mentality of a child.. her relationship with her parents is pretty sick.. they treat her like a little girl buy her off with sweets and money . I can't let another close friend kill himself.. what can I do? Thanks in advance..	
Summary A	Summary B
b""A close friend of mine says she will commit a suicide if she'll get kicked out of the University and her academic condition is pretty bad.. so she'll probably get kicked.... she says it not a threat or an idea, but a known fact that cannot be changed..""\r\n"	b""My friend thinks that suicide is the best solution for a difficult life. Two months ago, a close friend committed suicide, and her relationship with her parents is unhealthy. She's mentally immature and I'm desperate to help her.""\r\n"
Q1. Are there any obviously ungrammatical sentences in the summary, e.g., missing components, unrelated fragments or any other grammar-related problem that makes the text difficult to read?	
In Summary A	In Summary B
☐ Too many problems, the summary is impossible to read ☐ A fair amount of grammatical errors ☐ Some problems, but overall acceptable ☐ Minor grammar problems ☐ No noticeable grammatical problems	☐ Too many problems, the summary is impossible to read ☐ A fair amount of grammatical errors ☐ Some problems, but overall acceptable ☐ Minor grammar problems ☐ No noticeable grammatical problems

Figure A1. A sample of the MTurk study page.